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# Hardware-Assisted Dynamic Heat Redistribution Using PCM and Logic Control for Efficient Thermal Management in Advanced VLSI

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# INTRODUCTION

- Advanced VLSI systems operate at high transistor density, leading to increased power density and thermal hotspots
- Hotspots negatively impact performance, reliability, and device lifespan
- Existing thermal techniques (DVFS, cooling systems) suffer from latency, power overhead, and limited adaptability
- A real-time, hardware-based thermal management approach using PCM and logic control is required

# PROBLEM STATEMENT

- Localized hotspots cause performance degradation, leakage increase, and accelerated aging.
- Passive methods (PCM alone) lack control over heat redistribution.
- Software-based thermal management introduces delay and inefficient response.
- Absence of a low-latency, adaptive on-chip thermal control mechanism.

# EXISTING SOLUTION

- DVFS and software-based thermal control introduce latency and reduce overall system performance during operation.
- Heat sinks and conventional cooling techniques operate mainly at the package level and do not provide effective on-chip thermal management.
- Passive PCM cooling can absorb heat temporarily but suffers from saturation and lacks control over heat redistribution.
- Existing approaches do not provide real-time, localized, and adaptive thermal redistribution within the chip.

## PROPOSED FRAMEWORK

- The proposed framework introduces a PCM-assisted, logic-controlled thermal management system for efficient heat regulation in VLSI circuits.
- It integrates thermal sensing, hardware-based control, and heat redistribution mechanisms to manage localized hotspots.
- The framework enables real-time, low-latency thermal response without relying on software-based control.

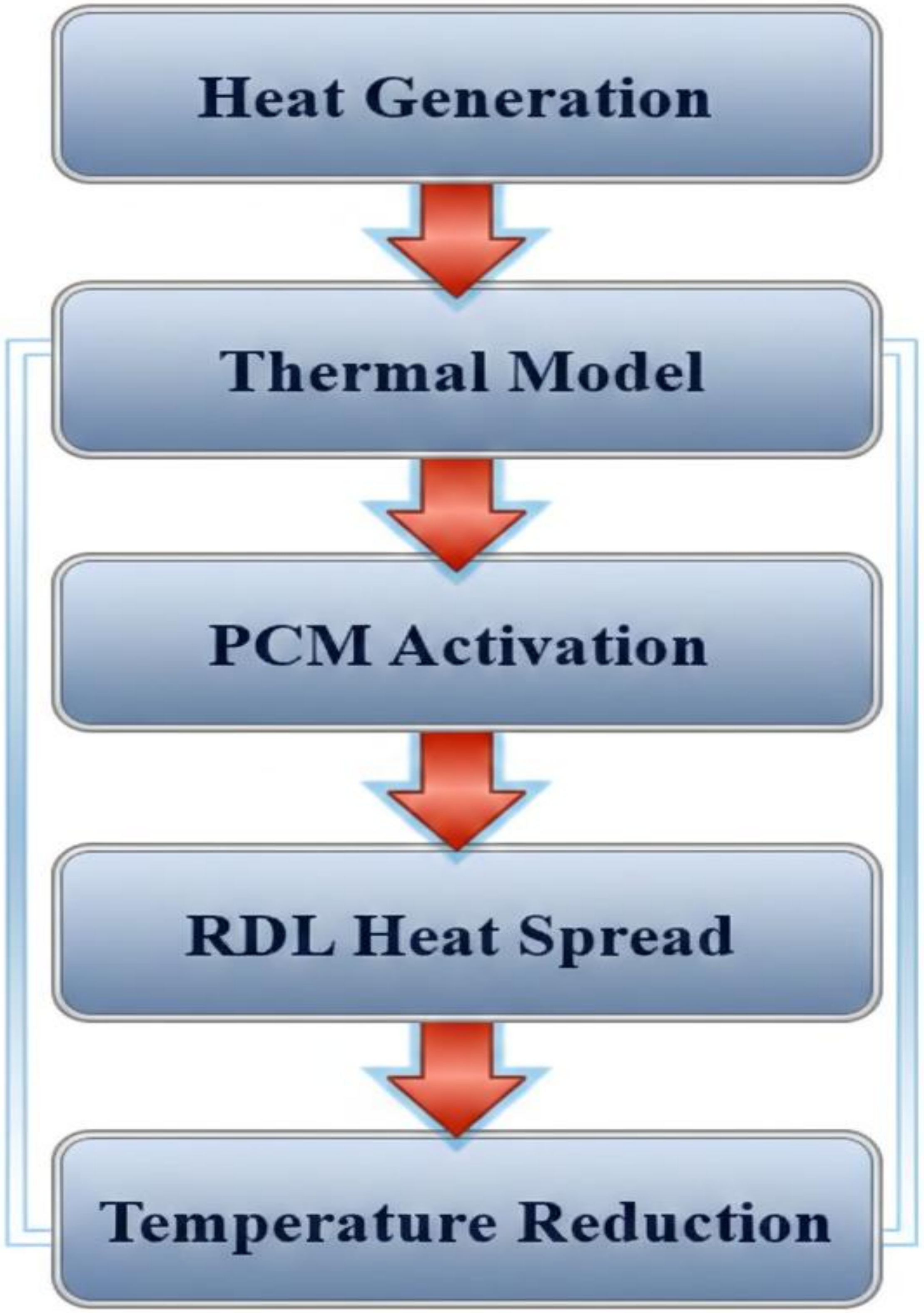



Figure 1.1. High-level operational flow of the PCM-assisted thermal management system

## SYSTEM OVERVIEW

- The system consists of thermal sensors, a logic control unit, PCM regions, and a redistribution layer forming a closed-loop architecture.
- Temperature data from sensors is continuously monitored and processed by the control unit to detect hotspot conditions.
- When a threshold is exceeded, PCM absorbs heat and the redistribution layer spreads it, ensuring dynamic thermal stabilization.

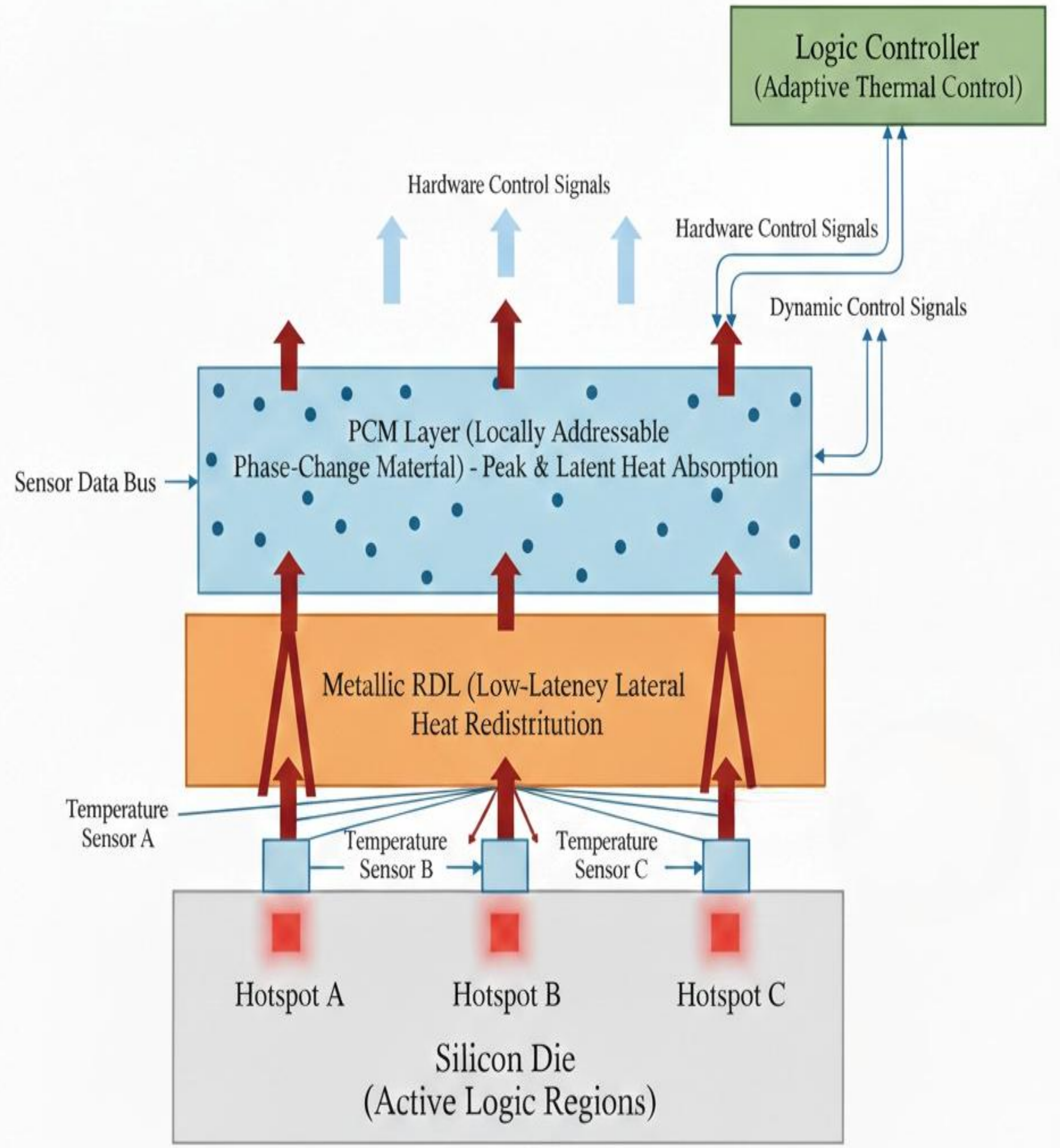



**Figure 1.2. On-chip PCM-assisted architecture with logic-controlled heat redistribution**

# SIMULATION & VALIDATION

- The proposed system is validated using multiphysics thermal simulation (ElmerSolver) along with control logic verification.
- Simulation models include heat conduction, PCM phase transition, and dynamic heat redistribution across the chip.
- Results confirm effective hotspot detection and real-time activation of control signals for thermal regulation.

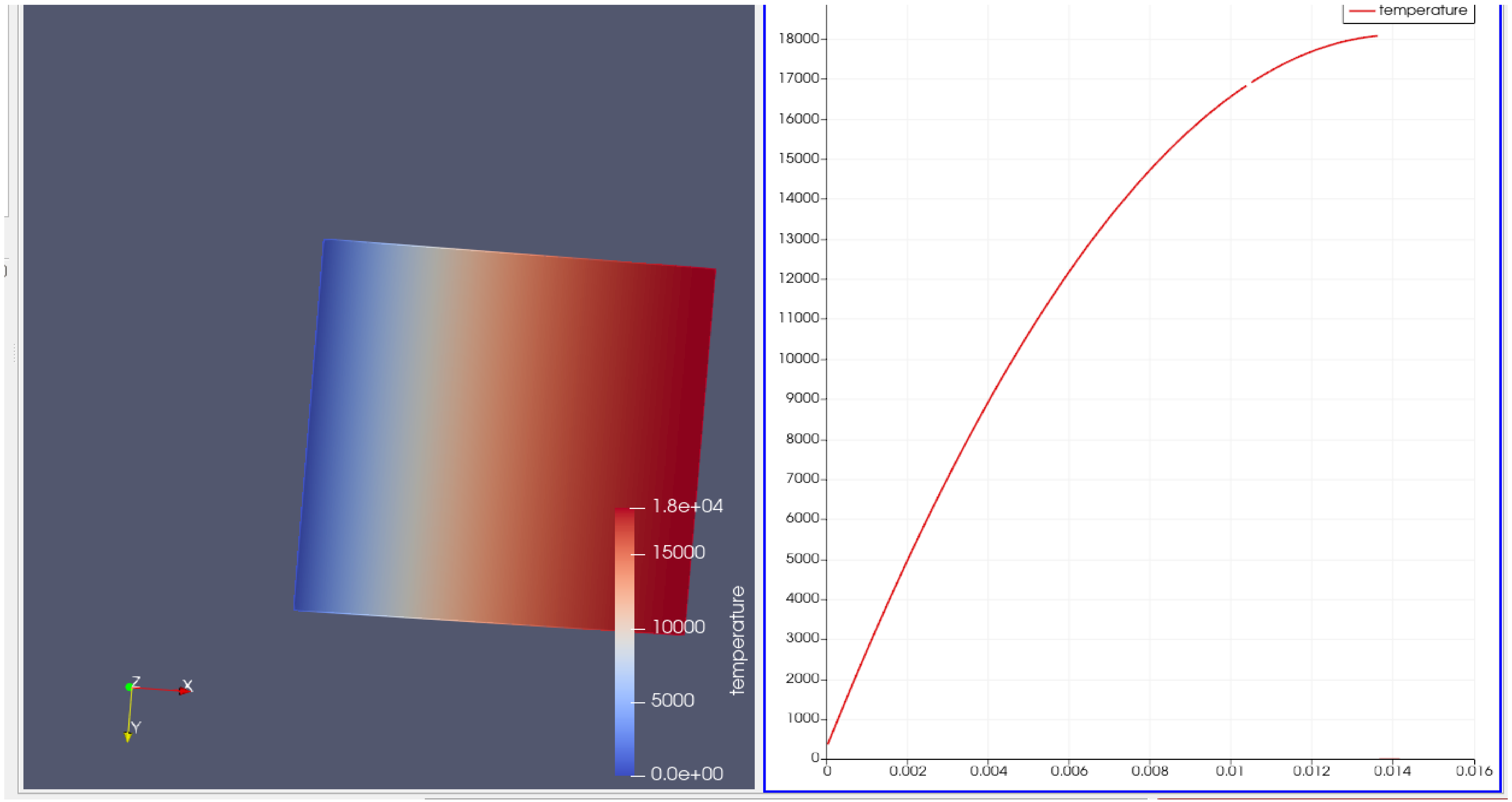


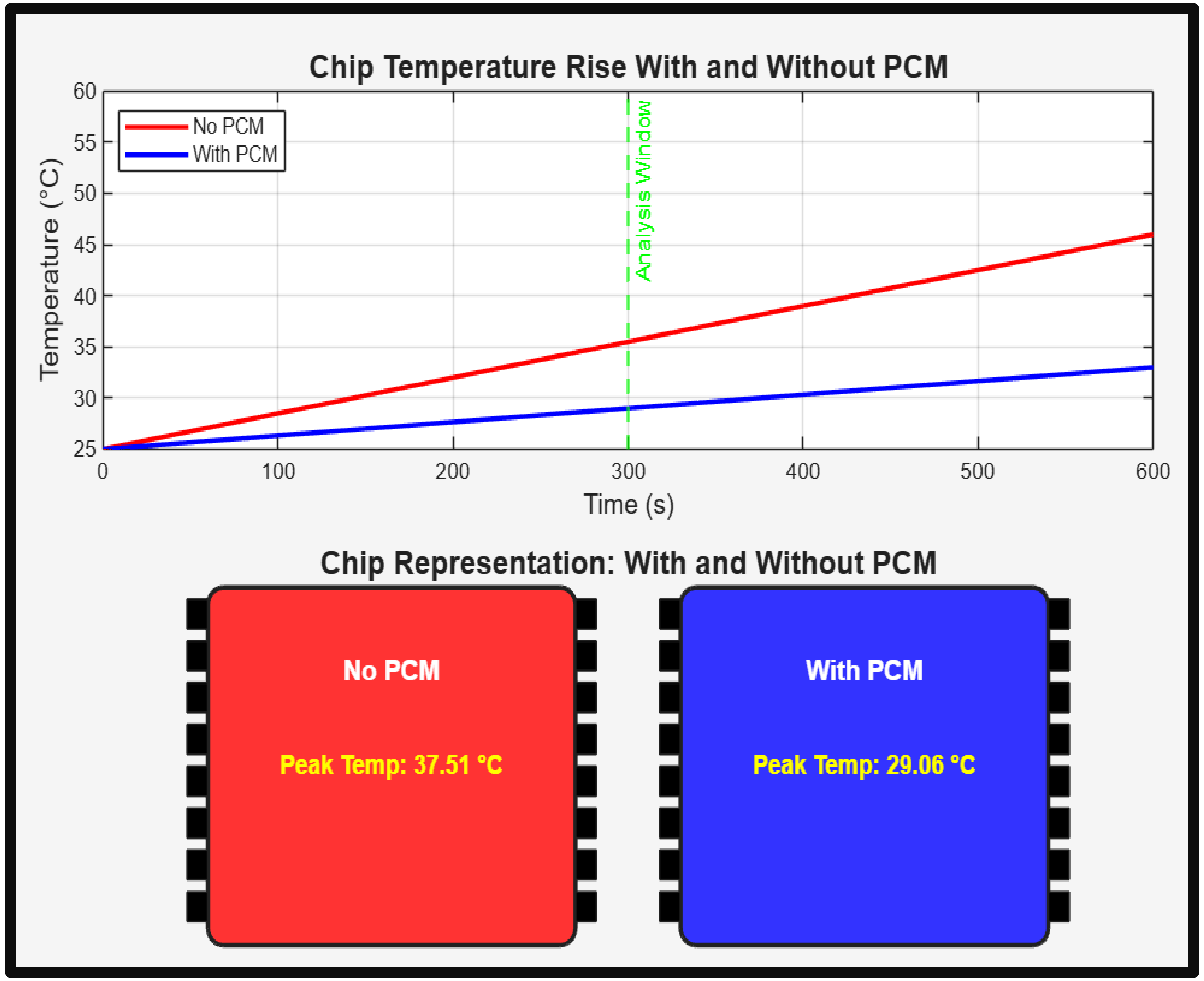

**Figure 1.3. Thermal distribution and simulation response of the proposed system**

## RESULT & DISCUSSION

- The proposed approach achieves significant reduction in peak hotspot temperature and improved thermal uniformity.

- Dynamic response analysis shows fast stabilization of temperature under varying thermal conditions.

- Overall system demonstrates efficient, low-latency thermal management compared to conventional methods.



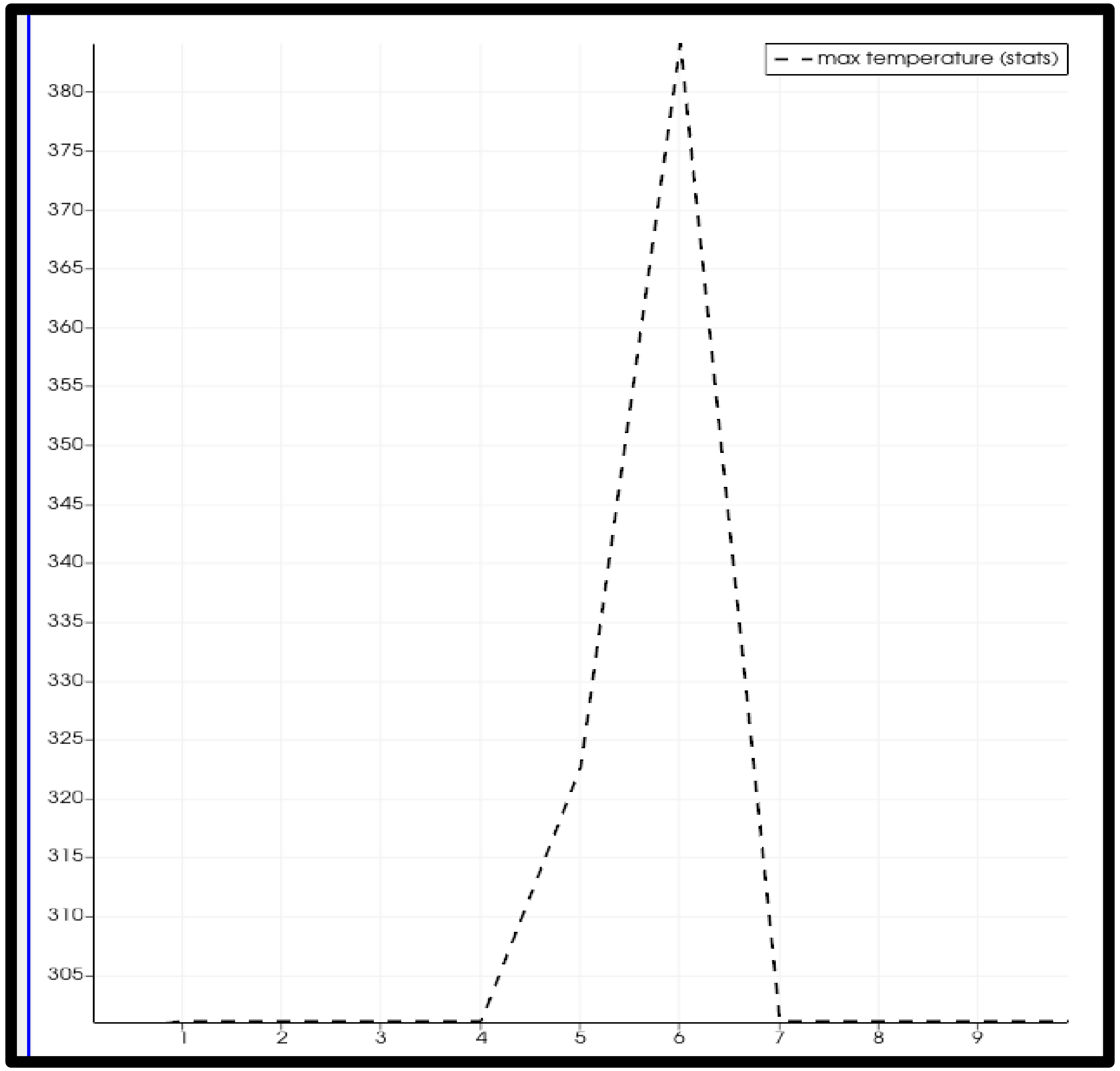




**Figure 1.4. Temperature comparison with and without PCM showing reduced peak temperature**

**Figure 1.5. Dynamic hotspot temperature response over time**

# COMPARISON WITH EXISTING THERMAL MANAGEMENT TECHNIQUES

| Technique | Response Time | Control Type | Efficiency | Limitation |
|---|---|---|---|---|
| DVFS | High latency | Software-based | Moderate | Performance loss |
| Heat Sink / Cooling | Slow | Passive | Moderate | No on-chip control |
| PCM (Passive) | Medium | Passive | Good | Saturation issue |
| Software Thermal Control | High latency | Software | Moderate | Delayed response |
| Proposed System | Near-zero latency | Hardware-based | High | Needs hardware validation |

# CONCLUSION & FUTURE WORK

- The proposed framework achieves real-time hotspot mitigation using PCM-assisted logic control with near-zero latency.
- It improves thermal stability, reduces peak temperature, and enhances overall system reliability.
- The system is scalable, energy-efficient, and compatible with CMOS fabrication processes.
- Future work includes hardware implementation, FPGA prototyping, and advanced PCM material optimization.

# THANK YOU!